Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/28/2013
PI Name:	Delp, Michael Ph.D.		
Project Title:	Regional Arterial Remodeling Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	mdelp@fsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-214-1195
Organization Name:	Florida State University		
PI Address 1:	College of Human Sciences		
PI Address 2:	242 Sandels Building, 120 Convocation Way		
PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 , STS-133 , STS-135 , Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX09AP06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tail suspension or hindlimb unloading of rodents has been used as a ground-based model of microgravity because tail suspension, like actual microgravity, 1) induces a headward shift of fluids and 2) diminishes the contractile activity of hindlimb skeletal muscles that are normally involved in maintaining posture. It is now evident that the shift of fluid toward the head and the unloading of postural muscles alter the mechanical forces exerted on arteries, the vessels responsible for regulating tissue blood flow and arterial blood pressure. For example, the smooth muscle cells in the small arteries of hindlimb skeletal muscle atrophies with tail suspension, and results in a diminished ability of these vessels to vasoconstrict. The compromised ability to constrict reduces the ability of the rats to maintain normal blood pressure, and consequently, these animals are susceptible to abnormal reductions in arterial pressure when the body position of the animal is moved from a horizontal position to a 70o head-up position. Conversely, cerebral arteries in the head hypertrophy, which can limit brain blood flow and result in loss of consciousness during head-up tilt. The purpose of this proposal is to determine whether the fluid shifts and reduced activity of postural muscles that are induced by microgravity similarly alter rodent arterial vessel structure and some key cellular signaling pathways. In other words, does space flight result in smooth muscle atrophy of peripheral arteries and hypertrophy of cerebral arteries? This work will provide potentially important information about the mechanisms underlying the reduced fitness and the inability to properly regulate blood pressure of astronauts returning to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments will contribute toward attaining a better understanding of how fundamental biological systems, such as the cardiovascular system, respond to the microgravity environment. The detailed study of the resulting vascular adaptations triggered by microgravity will yield essential information on the basic physiological responses of individual blood vessels involved in blood flow and pressure regulation. Further, analysis of cerebral arteries will provide potentially important information about the visual impairment of astronauts induced by microgravity This information will also support the development of treatments or countermeasures to improve crew health and performance following their return to a gravitational environment.
Task Progress:	From the shuttle studies all of the physiological and structural analyses of mesenteric, gastrocnemius, and cerebral resistance arteries are complete, and manuscripts containing these data are published or in press (see Bibliography). Data collection for the Bion M1 studies is ongoing, with studies from AEM control animals to be conducted later in 2013. Once these studies are completed data analysis can commence.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Stabley JN, Dominguez JM 2nd, Dominguez CE, Mora Solis FR, Ahlgren J, Behnke BJ, Muller-Delp JM, Delp MD. "Spaceflight reduces vasoconstrictor responsiveness of skeletal muscle resistance arteries in mice." J Appl Physiol. 2012 Nov;113(9):1439-45. http://dx.doi.org/10.1152/japplphysiol.00772.2012 ; PubMed PMID: 2298424 , Nov-2012
Articles in Peer-reviewed Journals	Behnke BJ, Stabley JN, McCullough DJ, Davis RT 3rd, Dominguez JM 2nd, Muller-Delp JM, Delp MD. "Effects of spaceflight and ground recovery on mesenteric artery and vein constrictor properties in mice." FASEB J. 2013 Jan;27(1):399-409. Epub 2012 Oct 25. http://dx.doi.org/10.1096/fj.12-218503 ; PubMed PMID: 23099650 , Jan-2013
Articles in Peer-reviewed Journals	Taylor CR, Hanna M, Behnke BJ, Stabley JN, McCullough DJ, Davis RT 3rd, Ghosh P, Papadopoulos A, Muller-Delp JM, Delp MD. "Spaceflight-induced alterations in cerebral artery vasoconstrictor, mechanical, and structural properties: Implications for elevated cerebral perfusion and intracranial pressure." FASEB J. 2013 Jun;27(6):2282-92. Epub 2013 Mar 1. http://dx.doi.org/10.1096/fj.12-222687 ; PubMed PMID: 23457215 ; PubMed Central PMCID: PMC3659353 , Jun-2013